Peter Mühlbacher

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I build and evaluate AI agents that forecast and do research. Before that, forecasting research at Metaculus and a PhD in probability theory.

EXPERIENCE

FutureSearch Oct 2023 – present
VP of Research (since Oct 2026); Founding Research Scientist (2023–2026)

- Build LLM agents for forecasting and web research, and the statistics used to evaluate them.
- Co-author of FutureSearch's public benchmarks for research and forecasting agents (below).

Metaculus Sep 2022 – Oct 2023
Research Scientist

- Researched methods for aggregating probabilistic forecasts.
- Built a "trust score" that predicts how reliable a crowd forecast is.

EDUCATION

University of Warwick 2022
PhD Mathematics. Thesis: *Probabilistic Methods for Quantum Spin Systems*

University of Cambridge 2017
MASt Mathematics (Part III)

University of Vienna 2016
BSc Mathematics

SELECTED PUBLICATIONS

- *Bench to the Future: A Pastcasting Benchmark for Forecasting Agents.* FutureSearch, arXiv:2506.21558, 2025.
- *Deep Research Bench: Evaluating AI Web Research Agents.* FutureSearch, arXiv:2506.06287, 2025.
- *Contra Papers Claiming Superhuman AI Forecasting.* With N. Bosse, L. Phillips, D. Schwarz. LessWrong, 2024.
- *Exploring Metaculus's AI Track Record.* With P. Scoblic, Metaculus, 2023.
- *Dimerization in Quantum Spin Chains with O(n) Symmetry.* With J. E. Björnberg, B. Nachtergaele, D. Ueltschi. Communications in Mathematical Physics 387, 2021.
- Further papers in probability and mathematical physics in ALEA (2021) and Random Matrices: Theory and Applications (2019).

AWARDS

- OeAD research scholarship (€2,200) and paid research internship at IST Austria (3% acceptance rate), 2016–2018.
- Dr. Hans Riegel Award (€600, 2012), for a paper on elliptic curves in public-key cryptography.

OTHER

- Languages: German (native), English (fluent); basic French, Russian, Chinese.
- Former competitive gymnast.